

1. Understanding the concepts

Mark the correct statements by ■□ and the incorrect ones by □■. You learn from your mistakes.

- The empty set $\emptyset$ can be made into a vector space $(\emptyset, \oplus, \odot)$ by an appropriate choice of $\oplus$ and $\odot$.
- The maps $\oplus$ and $\odot$ of a vector space $(V, \oplus, \odot)$ map two elements of V to one element of V .
- The set V underlying a vector space $(V, \oplus, \odot)$ must contain at least one element.
- If both $(V, \oplus, \odot)$ and $(V, \boxplus, \boxdot)$ are vector spaces, then they are inevitably identical.
- If both $(V, \oplus, \odot)$ and $(W, \oplus, \odot)$ are vector spaces, then inevitably $V = W$.
- If $(V, \oplus, \odot)$ is a vector space, a bi-linear map $\mu : V \times V \rightarrow V$ satisfies $\mu(\lambda \odot v, \lambda \odot v') = \lambda \odot \mu(v, v')$.
- Every vector space $(V, \oplus, \odot)$ carries an inner product defined by $\odot$.
- A (p, q) -tensor T is a multi-linear map $T : \underbrace{V \times \cdots \times V}_p \times \underbrace{V^* \times \cdots \times V^*}_q \rightarrow V$.

2. Mastering the formalism

Mastership is in the basics. Work hard to find the required short proofs. This exercise will pay off.

- (a) Prove that a vector space $(V, \oplus, \odot)$ has only one neutral element O .
- (b) Prove that for every linear map $\xi : V \rightarrow W$ between vector spaces $(V, \oplus, \odot)$ and $(W, \boxplus, \boxdot)$

$$\xi(O_V) = O_W,$$

where O_V denotes the neutral element in V and O_W the neutral element in W .

- (c) Consider three vector spaces $(V, \oplus, \odot)$, $(W, \boxplus, \boxdot)$ and $(U, \boxplus, \boxdot)$ and two linear maps

$$\varphi : V \rightarrow W \quad \text{and} \quad \psi : W \rightarrow U.$$

Prove that their composition, defined by

$$\psi \circ \varphi : V \rightarrow U, \quad (\psi \circ \varphi)(v) := \psi(\varphi(v))$$

for any $v \in V$, is again a linear map.

- (d) Prove, from our definition of an inner product g on a vector space $(V, \oplus, \odot)$, that

$$g(v, v) = 0 \quad \text{if and only if} \quad v = O$$

and that the inequality

$$|g(v, v')|^2 \leq g(v, v)g(v', v')$$

holds for all $v, v' \in V$.

- (e) Prove that for any linear map $\varphi : V \rightarrow W$ between vector spaces $(V, \oplus, \odot)$ and $(W, \boxplus, \boxdot)$, the set

$$\ker \varphi := \{v \in V \mid \varphi(v) = O_W\} \quad (\text{the so-called } \textit{kernel} \text{ of the map})$$

is a vector subspace of V , which is the case by definition if for all $a, b \in \ker \varphi$ and $\lambda \in \mathbb{R}$ also

$$\lambda \odot a \in \ker \varphi \quad \text{and} \quad a \oplus b \in \ker \varphi.$$

- (f) Let $(V, \oplus, \odot)$ be a vector space and $\varphi : V \rightarrow V^*$ and $\psi : V \rightarrow V$ be linear maps. Prove that

$$\pi : V \times V \rightarrow \mathbb{R}, \quad \pi(v, v') := \varphi(v)(\psi(v'))$$

is a $(0, 2)$ -tensor.